

Pandas Complete Data Analyst Workflow

1. Import Libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Import Dataset:

```
df = pd.read_csv("file.csv")
```

3. Data Understanding:

```
df.head()
df.shape
df.info()
df.describe()
df.isnull().sum()
df.duplicated().sum()
```

4. Primary Key Handling:

Check:

```
df["Order_ID"].is_unique
```

IF PK EXISTS:

- Check duplicate
- Remove duplicate

```
df.drop_duplicates(subset="Order_ID")
```

IF PK NOT EXISTS:

Create ID

```
df["ID"]=range(1,len(df)+1)
```

5. Data Cleaning:

Remove Duplicate:

```
df.drop_duplicates()
```

Missing Values:

```
df.fillna()
```

Change Datatype:

```
astype()
```

```
pd.to_datetime()
```

Remove Spaces:

```
str.strip()
```

6. Data Transformation:

Create New Columns:

Profit:

```
df["Profit"]=df["Sales"]-df["Cost"]
```

Month:

```
df["Month"]=df["Date"].dt.month
```

7. Create Dimension Tables:

Dim_Customer:

- Drop duplicate customers
- Create Customer_ID

Dim_Product:

- Drop duplicate products
- Create Product_ID

Dim_Date:

- Create Date table
- Add Year Month Quarter

8. Data Analysis:

Total:

sum()

Average:

mean()

Sorting:

sort_values()

Filtering:

conditions

Grouping:

groupby()

9. Visualization:

Bar Chart:

`sns.barplot()`

Line Chart:

`sns.lineplot()`

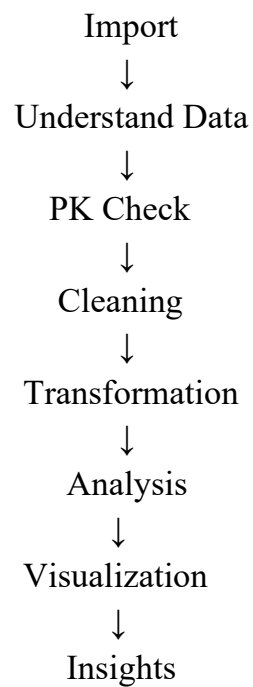
Pie Chart:

`plot(kind="pie")`

Heatmap:

`sns.heatmap()`

FINAL FLOW:



Sample Dashboard:

